

# Mathieu Petitbois

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## SUMMARY

I am a final-year PhD candidate at Ubisoft La Forge and LERIA advised by [Ludovic Denoyer](#), [Sylvain Lamprier](#), and [Rémy Portelas](#). My research focuses on **Deep Reinforcement Learning**, **Imitation Learning**, and **Stylized Behavior Modeling** for physics-based characters and autonomous agents in complex interactive environments. I am currently seeking **Research Scientist** and **Postdoctoral** opportunities, with a focus on **Generative Modeling**, **Large Language Models**, **Sequential Decision-Making**, and **Embodied AI**.

## WORK EXPERIENCE

### Ubisoft La Forge

Bordeaux, France

*Research Assistant*

*June 2023 – May 2026*

- Developed and integrated generative models (Transformers, Diffusion, VAEs, VLMs) to bridge generative modeling and sequential decision-making in complex environments (Godot, Atari, MuJoCo, Newton Physics).
- Collaborated to integrate research outcomes into game development through scientific consulting.
- Authored, published, and presented research findings at international conferences and venues.

### Ubisoft La Forge

Bordeaux, France

*Research Engineer*

*Oct. 2022 – May 2023*

- Designed and implemented a library to extract behavioral diversity from player data, enabling the learning of diverse, controllable policies for behavior personalization.

### Ubisoft La Forge

Bordeaux, France

*Research Intern*

*Apr. 2022 – Sep. 2022*

- Implemented and evaluated offline and online RL methods in simulated environments using MuJoCo (physics simulator) and Panda3D (game engine).
- Demonstrated that offline RL improves sample efficiency and reduces training time given sufficient data, contributing to a shift in company direction from online RL to data-driven offline RL for policy learning in games.

### ENSTA Paris, U2IS

Palaiseau, France

*Research Intern*

*June 2021 – August 2021*

- Reimplemented the model-based approach from the World Models paper (Ha & Schmidhuber, NeurIPS 2018) in PyTorch.

## EDUCATION

### Ubisoft La Forge & Université d'Angers, LERIA

Bordeaux & Angers, France

*PhD in Deep Reinforcement Learning for Stylized Behavior Modeling*

*June 2023 – May 2026*

- **Supervision:** [Ludovic Denoyer](#), [Sylvain Lamprier](#), and [Rémy Portelas](#).

### Institut Polytechnique de Paris

Palaiseau, France

*M.Sc. in Data Science, Highest Honors*

*Sep. 2021 – Sep. 2022*

- **Main Courses:** Deep Learning, Reinforcement Learning, Computer Vision.

### ENSTA Paris

Palaiseau, France

*Master's Degree in Engineering*

*Sep. 2019 – Sep. 2022*

- **Main Courses:** Dynamic Programming, Logic Programming, Linear Programming, Genetic Algorithms, Probability Theory, Optimization, Applied Statistics, Statistical Learning, Control Theory, Bayesian Filtering.

## TECHNICAL SKILLS

**Languages:** French (Native), English (C1/C2, TOEIC: 940/990), German (B1/B2)

**Programming Languages:** Python, LaTeX

**Libraries & Frameworks:** PyTorch, JAX, NumPy, Pandas, Hydra, Transformers, vLLM, MuJoCo, Newton Physics

**Tools:** Git, Docker, VS Code, GitHub Copilot, Slurm

## PUBLICATIONS

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- Offline Reinforcement Learning of High-Quality Behaviors Under Robust Style Alignment** 2026  
M. Petitbois, R. Portelas, S. Lamprier  
*ICML 2026 (Spotlight)*  
*Agents in the Wild: Safety, Security, and Beyond Workshop at ICLR 2026*  
*Principled Design for Trustworthy AI Workshop at ICLR 2026*
- Offline Goal-Conditioned Reinforcement Learning with Projective Quasimetric Planning** 2025  
A. Kobanda, W. Radji, M. Petitbois, O.A. Maillard, R. Portelas  
*EWRL 2025*
- Navigation with QPHIL: Quantizing Planner for Hierarchical Implicit Q-Learning** 2025  
M. Petitbois\*, A. Canesse\*, L. Denoyer, S. Lamprier, R. Portelas  
*IJCNN 2025*  
*7th Robot Learning Workshop at ICLR 2025*
- Offline Learning of Controllable Diverse Behaviors** 2025  
M. Petitbois, R. Portelas, S. Lamprier, L. Denoyer  
*Generative Models for Robot Learning Workshop at ICLR 2025*

\*Equal contribution